

Adhithyan Sakthivelu

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EDUCATION

Master of Science, Stanford University

Civil and Environmental Engineering - Atmosphere & Energy

Stanford, CA

June 2023

Bachelor of Technology, National Institute of Technology, Trichy

Instrumentation and Control Engineering

Trichy, India

June 2018

RESEARCH ROLES

Water & Energy Efficiency for the Environment (WE3) Lab

Research Assistant | PI: Prof. Meagan Maunder

Stanford, CA

April 2023 to Present

- Developing energy optimization algorithms and bidding strategies for water and industrial systems in demand response and energy markets

SLAC National Accelerator Lab & Post Road Foundation

Research Assistant | PI: Dr. David P. Chassin

Menlo Park, CA

April 2022 to Present

- Developing distributed energy resource (DER) bidding strategies and designing markets for transactive (peer-to-peer) energy system services (TESS)

PROJECT EXPERIENCES

Risk-aware bidding strategy for industrial participation in DR markets

Research Paper – In Preparation

Stanford, CA

June 2023 to Present

- Developed risk-aware optimal bidding strategies using **Monte Carlo methods**, **model predictive control**, and grid-search; implemented in case study wastewater treatment facilities
- Designed a parametric modular DR event simulator for stochastic analysis

Heuristic bidding strategies for DERs in TESS Markets

DOE Connected Communities Project

Menlo Park, CA

April 2022 to Present

- Formulated heuristic bidding strategies for HVAC, batteries, and EVs (**DERs**) based on customer preferences and market data for maximizing utility
- Validated the strategies with physical devices using EcoPort, Emporia and Tesla fleet API communication protocols

A real-time limit orderbook framework for TESS

Conference Paper – IEEE PESGM'23

Menlo Park, CA

July 2023

- Published a novel LOB market mechanism for price discovery and order-matching for real-time retail electricity markets; selected as one of the best conference papers

OTHER SELECTED PROJECTS

Optimized Operation of HVAC system with TOU electricity pricing and weather prediction data

Course Project

Stanford, CA

March 2022 to May 2022

- Created a physics-based (ETP) MPC model for electric heat pumps that minimizes the total cost while maintaining comfort for the customers given the electricity retail price and weather prediction

An Energy-Efficient Ride-Sharing Algorithm Using Distributed Convex Optimization

Course Project

Stanford, CA

March 2022 to May 2022

- Developed a ride-sharing optimization with focus on energy-efficiency and decentralized decision-making (ADMM) agents for rider pickup and drop-off, as part of localized clusters of connected cars

Online US Wind Energy Atlas for potential windfarm development

Website - windenergyatlas.stanford.edu

Stanford, CA

May 2021 to Sep 2021

- Built a website tool for energy developers and policy makers to disseminate feasible areas of windfarm development in the United States using **ArcGIS online** platform; got presented at **Glasgow COP2**

WORK EXPERIENCE

Deloitte US India Consulting

Consultant

Bengaluru, India

August 2018 to August 2021

- Analyzed and processed raw customer data, transformed them to Salesforce compatible schema, and built **automated algorithm** for data housekeeping
- Customized and configured **Salesforce Lightning Platform** and Apttus CPQ managed package for Lead-to-Cash lifecycle

PUBLICATIONS

- A. Sreekumar*, A. Sakthivelu*, and L. Kiesling, “**Auction Theory and Device Bidding Functions for Transactive Energy Systems: A Review**,” Curr Sustainable Renewable Energy Rep, vol. 10, no. 3, pp. 102–111, Sep. 2023, doi: [10.1007/s40518-023-00217-2](https://doi.org/10.1007/s40518-023-00217-2) (*equal contribution)
- A. Sreekumar, A. Sakthivelu, R. Baltaduonis, L. Kiesling, S. Hoedl, and D. P. Chassin, “**A Real-Time Limit Order Book as a Market Mechanism for Transactive Energy Systems**,” in 2023 IEEE Power & Energy Society General Meeting (PESGM), Jul. 2023, pp. 1–5. doi: [10.1109/PESGM52003.2023.10252385](https://doi.org/10.1109/PESGM52003.2023.10252385)

LEADERSHIP, COMMUNITY IMPACT, & INTERESTS

Energy Focus Area Lead

Stanford, CA

Stanford Doerr-IITB Workshop on Sustainability

June 2024

- Led four-day breakout room discussions with faculties from Stanford and IIT Bombay on collaborative research opportunities between the two institutions, resulting in a white paper

Research Mentor

Stanford, CA

SURGE & MUIR Fellowships

May 2024

- Mentored two undergraduate researchers through the SURGE fellowship on creating a machine-readable database for incentive-based DR programs, comprising of 140 programs across utilities in the US

Climbing Instructor

Stanford, CA

Stanford Outdoor and Recreation Center

January 2022

- Conducted indoor rock-climbing classes based on experiential learning, with focus on rope safety, injury prevention and physical efficiency skills

President

Trichy, India

Nakshatra, The Astronomy Club of NIT Trichy

2017 - 2018

- Managed the administrative activities of the club, and organized "Stardust," a two-day inter-mural astronomy fest with expert lectures, workshops, and stargazing sessions

Yoga Instructor

Stanford, CA

- Organized yoga sessions for community building, with focus on creative expression of movement and breathwork
Completed Diploma in Yoga Education and Therapy from Sri Paranjothi Yoga College, India

SKILLS, COURSEWORKS & AFFILIATIONS

- **Software Languages:** Python, Julia, MATLAB, Java, Apex (Salesforce), LaTeX
- **Mathematical Modeling and Optimization:** CVXpy, Pyomo, MATLAB, Convex.jl
- **Testing Frameworks:** Pytest, Unittest
- **Version Control & CI/CD:** Git, Github Actions
- **GIS:** QGIS, PyQGIS, GeoPandas, ArcGIS, ArcGIS online
- **Data Analysis and Database:** SQL Database, MS Excel – Advanced
- **Relevant Coursework:** Electricity Markets, Electricity Economics, Engineering Future Electricity Systems, Convex Optimization, Decision Making Under Uncertainty, Data Structures and Algorithms, Engineering Optimization, Digital Signal Processing, Product Design, Advanced Feedback Control
- **Affiliations:** IEEE Student Member (Since 2023)